Hearing physiology

Hearing is the sense by which we perceive sound.

Hearing

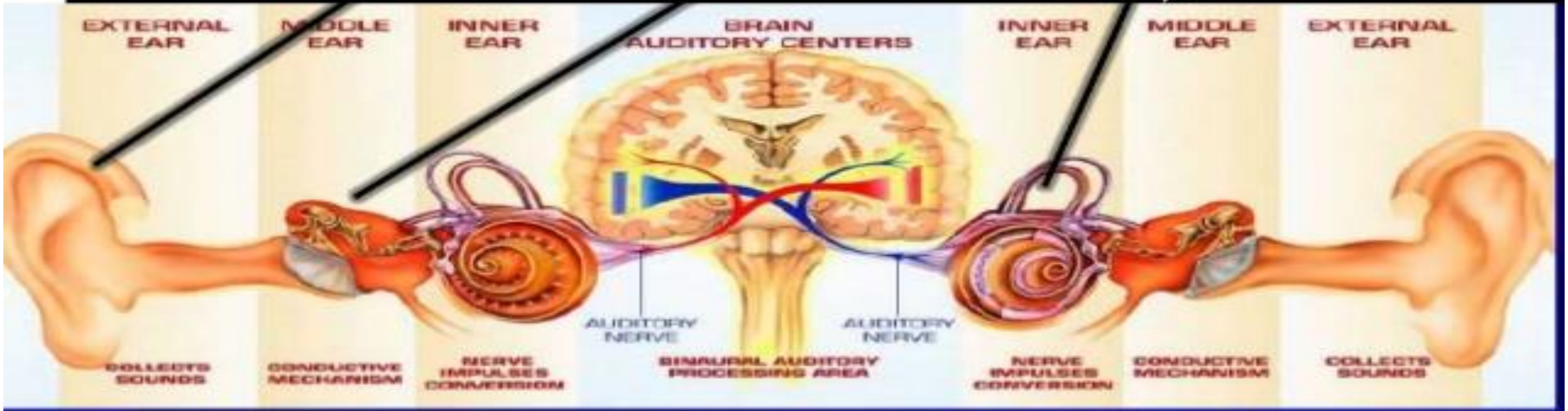
Hearing is the process by which the ear transforms sound vibrations in into nerve impulses that are conveyed to the brain

Where as

Listening is processing of sound waves in auditory area of brain to interpret the meanings of sound

EAR AS A TRANSDUCER

SOUNDENERGY → MECHANICAL ENERGY → ELECTRICAL ENERGY

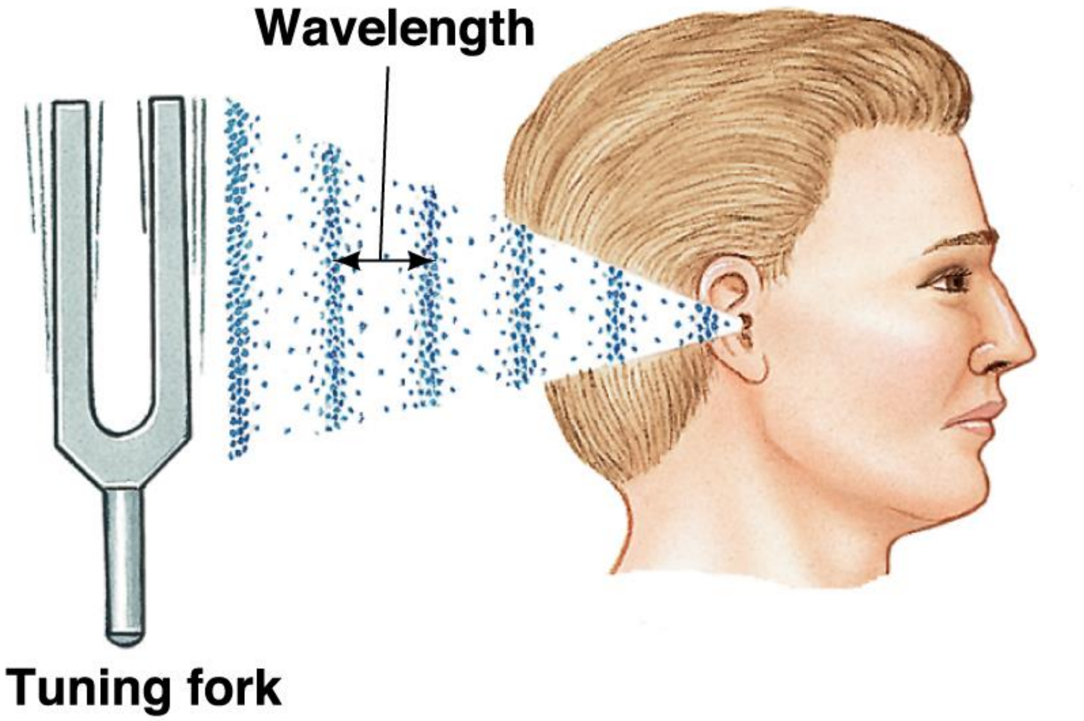


Schematic representation of normal hearing mechanism

Sound

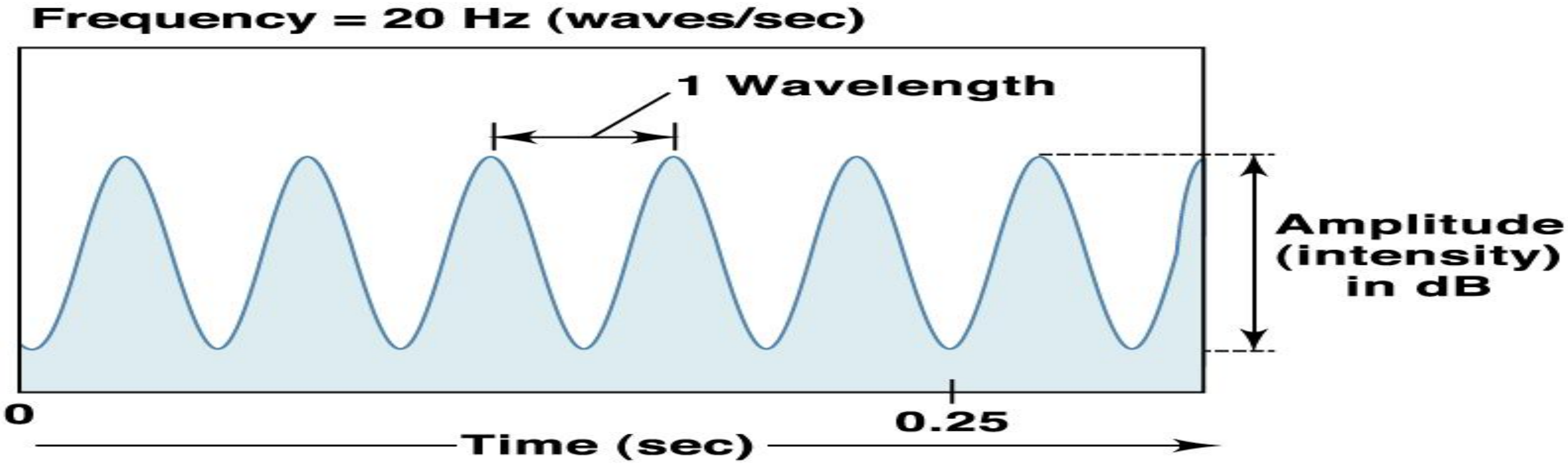
- Sound waves consist of alternate regions of compression & decompressions of air molecules
- Speed: 340 m/s through air

- Sounds cannot travel through a vacuum.



Properties

- **Pitch** = Frequency of waves arriving measured in hertz (Hz)
- The frequencies of sound a young person can hear are between 20 and 20,000 cycles/sec
- **Loudness** = Intensity of waves measured in decibels (dB)



Main Components of the Hearing

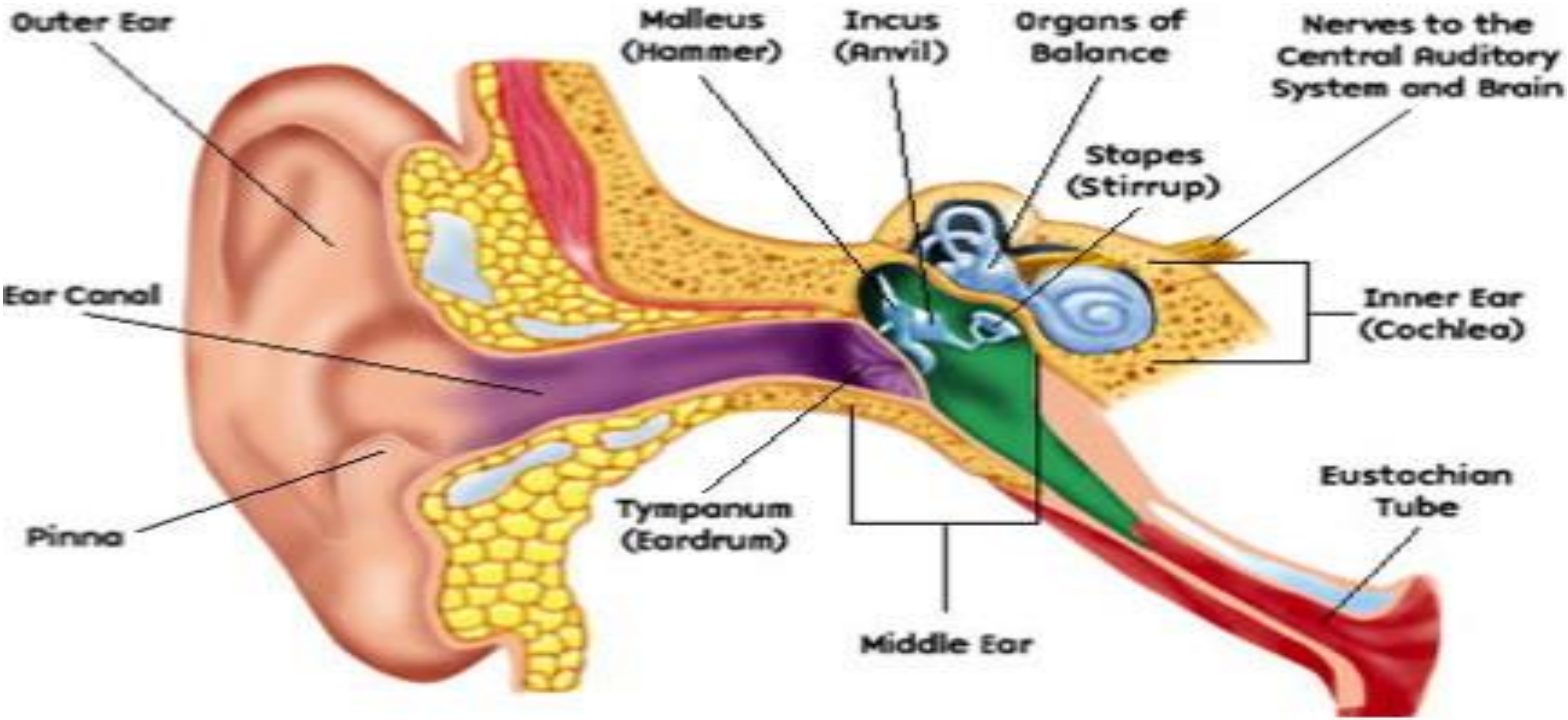
Peripheral

- – Outer ear
- – Middle ear
- – Inner ear
- – Auditory nerve

Central

- – Brainstem (medulla oblongata)
- – Midbrain
- – Cerebral cortex

Physiological anatomy of ear

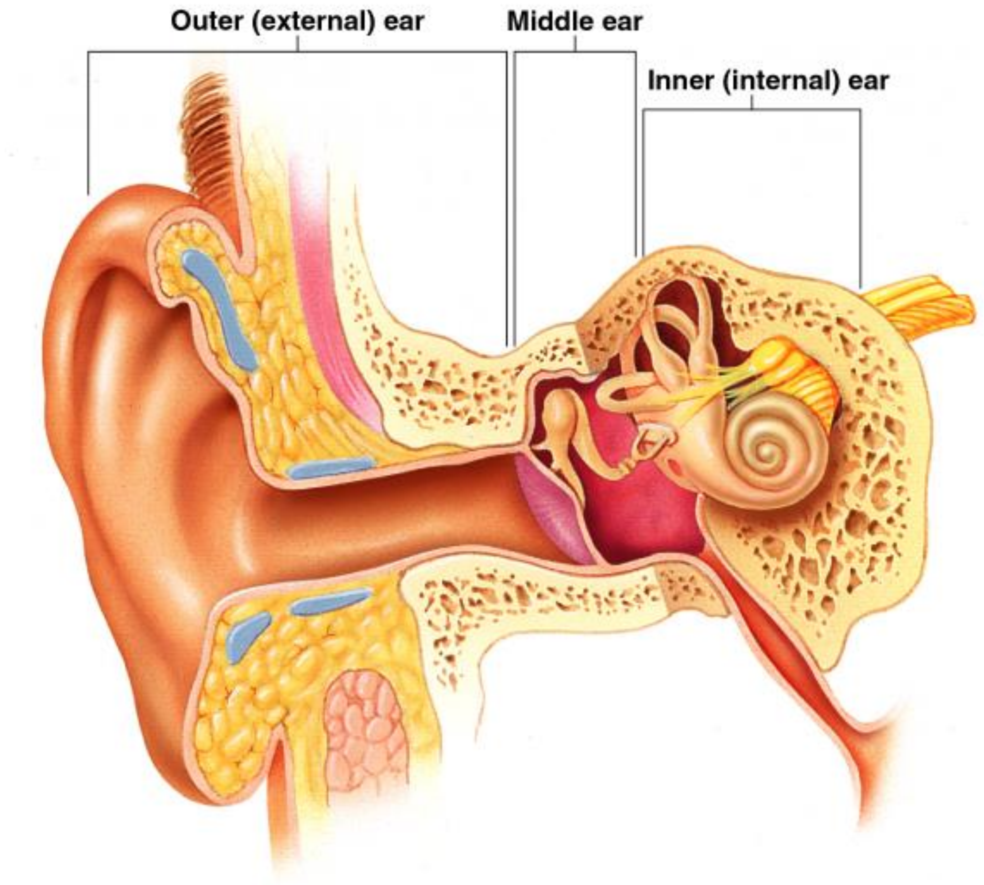


External ear

1. PINNA

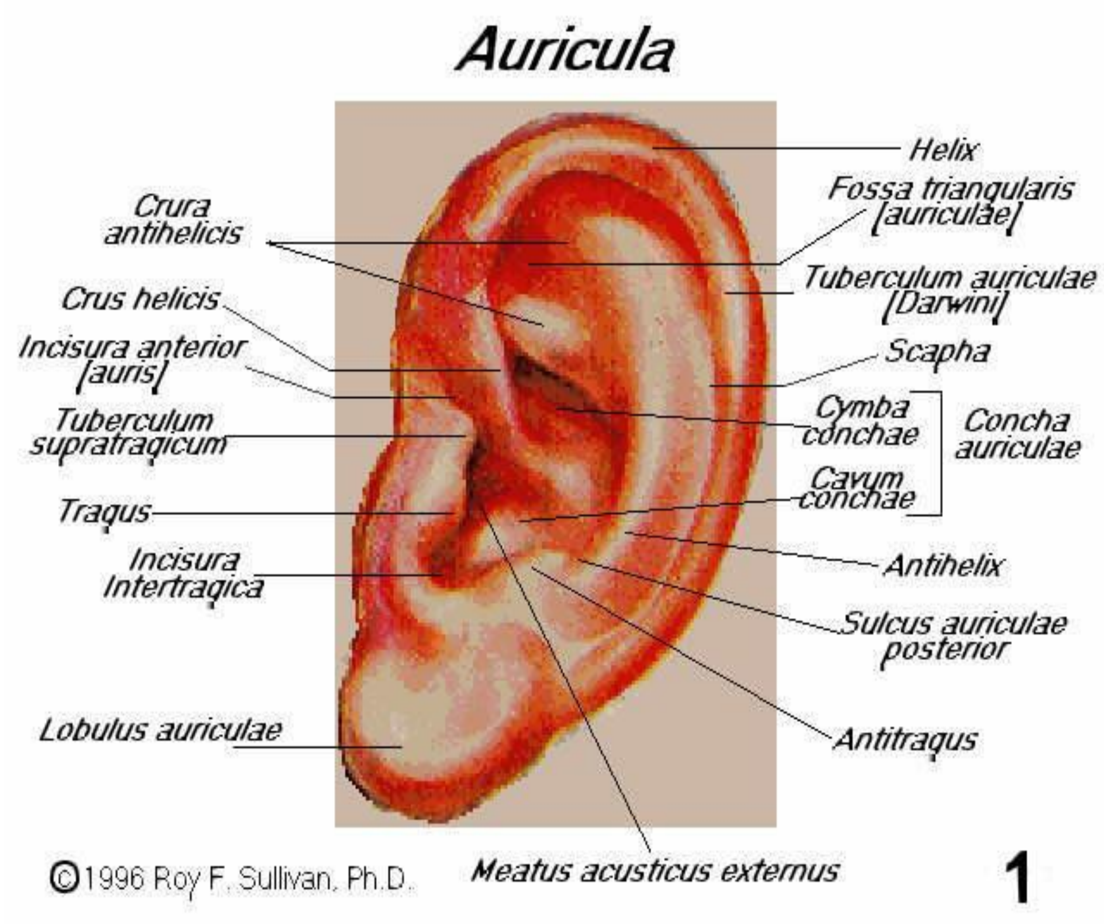
2. External auditory meatus/ canal

3. EARDRUM or TYMPANIC MEMBRANE:



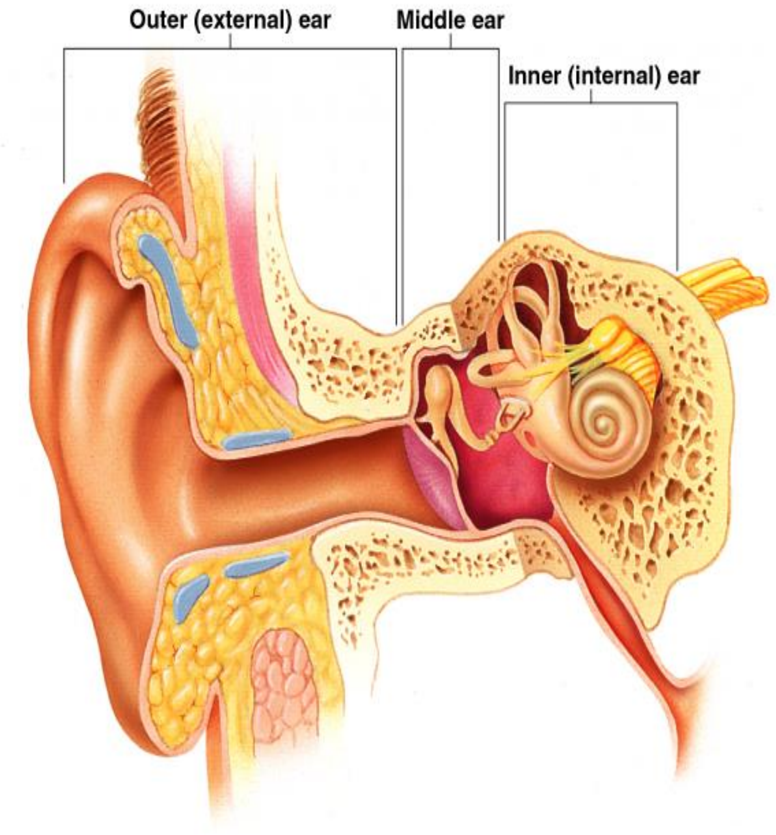
Structures of the Outer Ear

- **<u>Auricle (Pinna)</u>**
  - Collects sound and channels them through the air canal.
  - Helps in sound localization



The External Auditory Canal

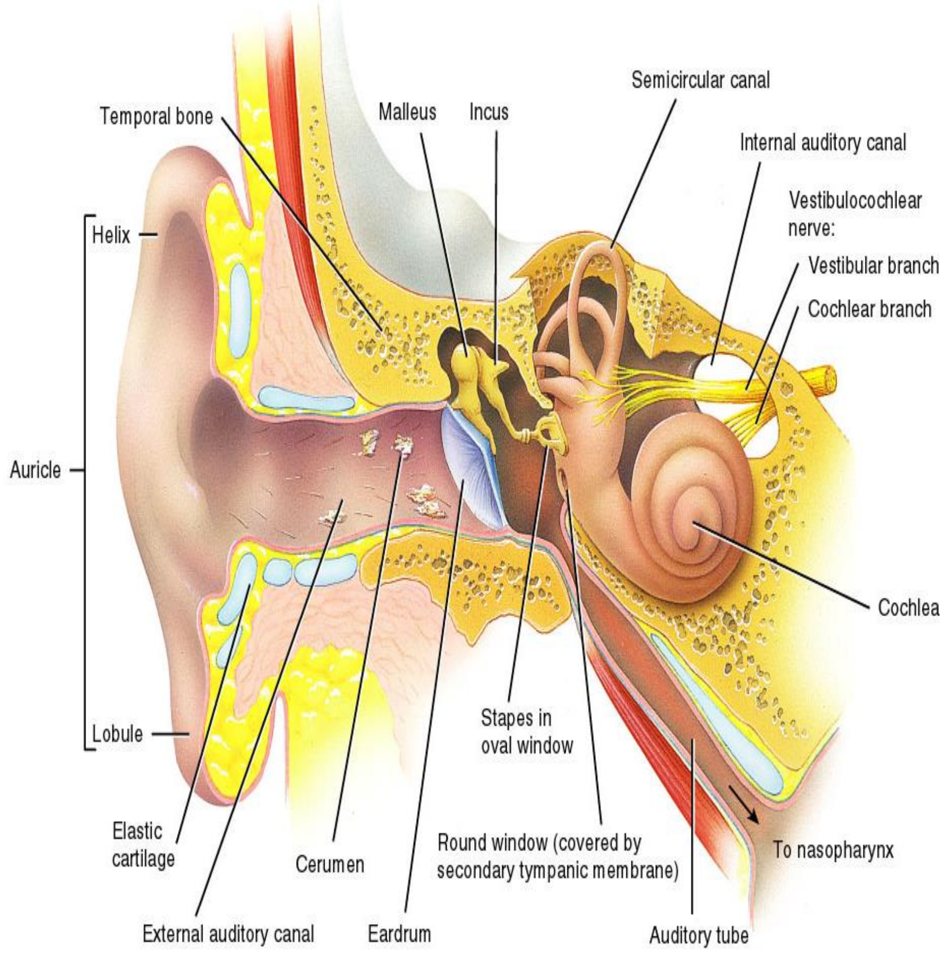
- Narrow chamber in the temporal bone
- Lined with skin
- Ceruminous (wax) glands are present
- Ends at the tympanic membrane **(eardrum)**



Defects of external ear produces conductive deafness

Tympanic Membrane

Changes sound energy into mechanical energy



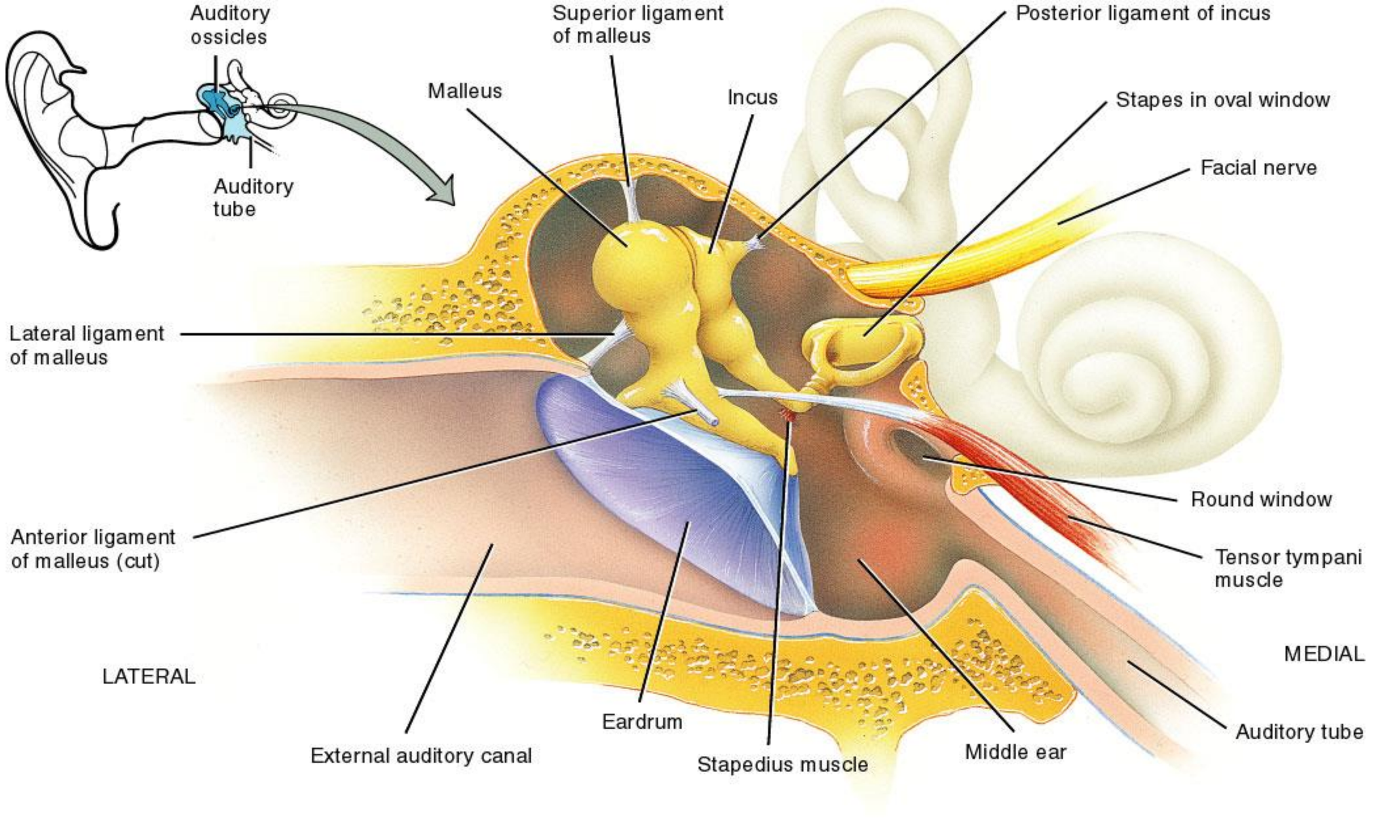
Frontal section through the right side of the skull showing the three principal regions of the ear

Boundary between external and middle ear

- Vibrates in response to sound
- vibrating slowly at low frequency and vibrate rapidly in response to high frequency

The Middle Ear or Tympanic Cavity

- Air-filled cavity within the temporal bone



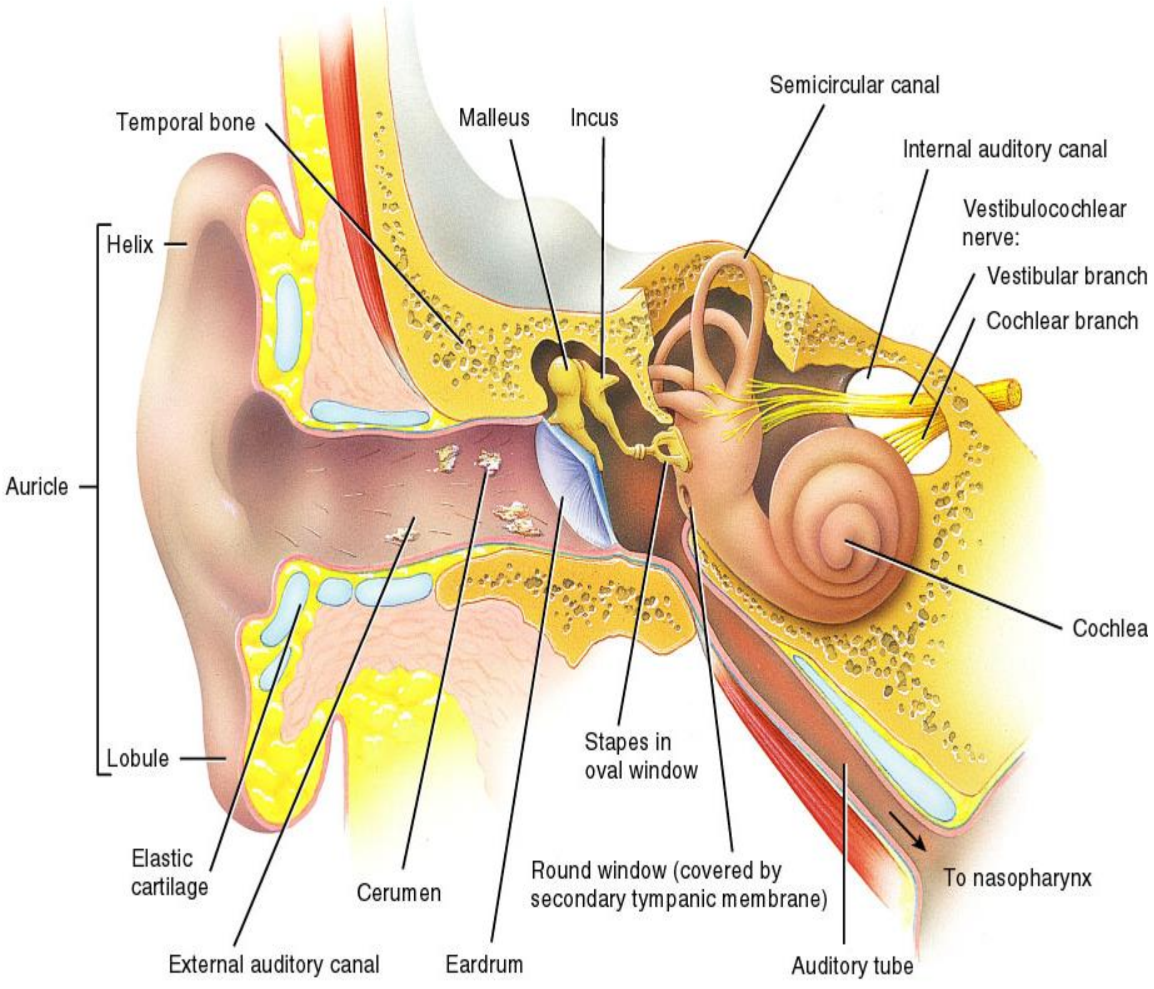
Frontal section showing location of auditory ossicles

16.18

The Ossicular Chain

- **A**: Malleus
- B: Incus
- C: Stapes
  - Ossicles are smallest bones in the body

  - Act as a lever system



Frontal section through the right side of the skull showing the three principal regions of the ear

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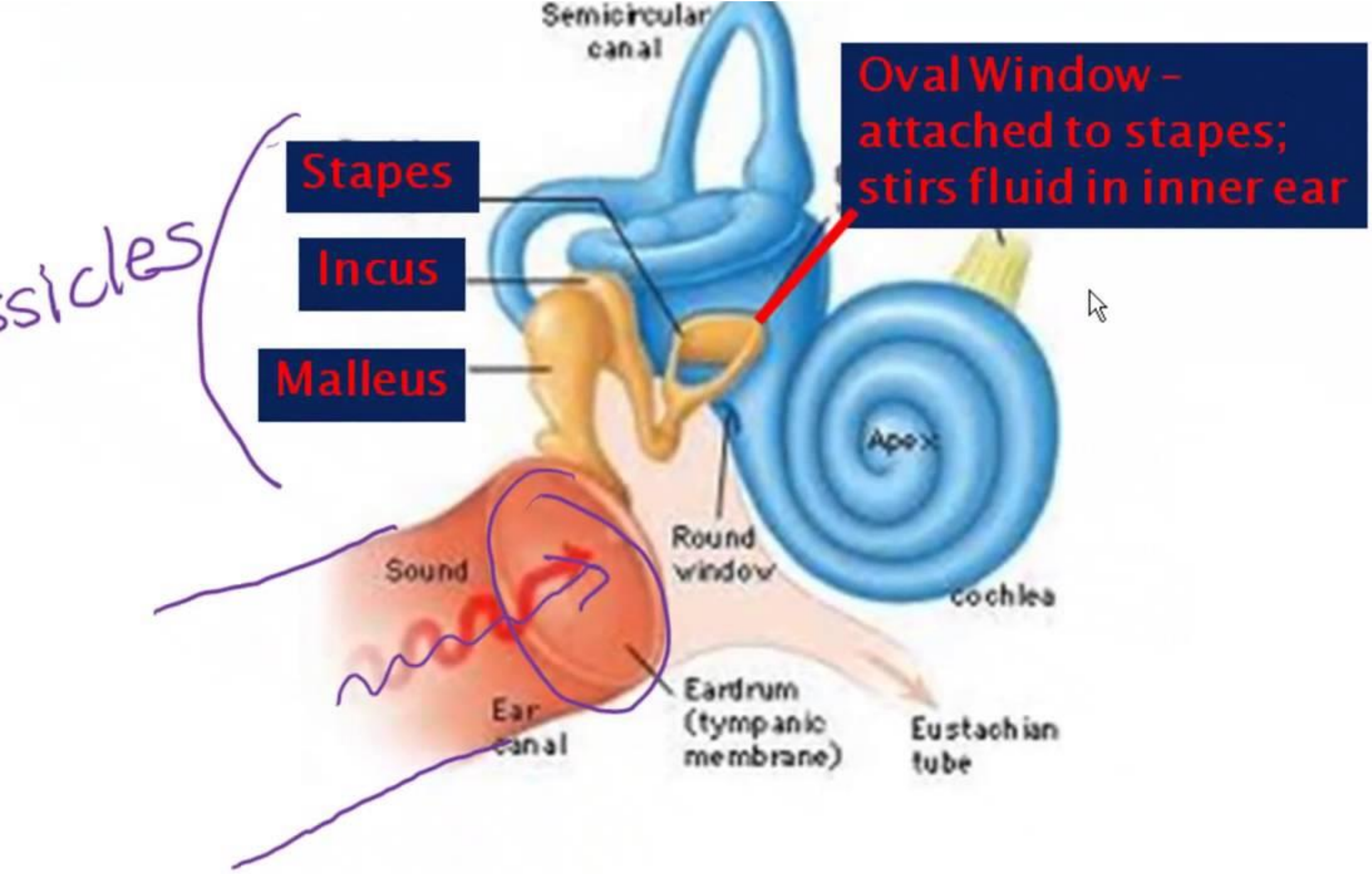


Semicircular canal
Oval Window – attached to stapes; stirs fluid in inner ear
Stapes
Incus
Malleus
ssicles
Apex
Round window
cochlea
Sound
Eardrum (tympanic membrane)
Ear canal
Eustachian tube

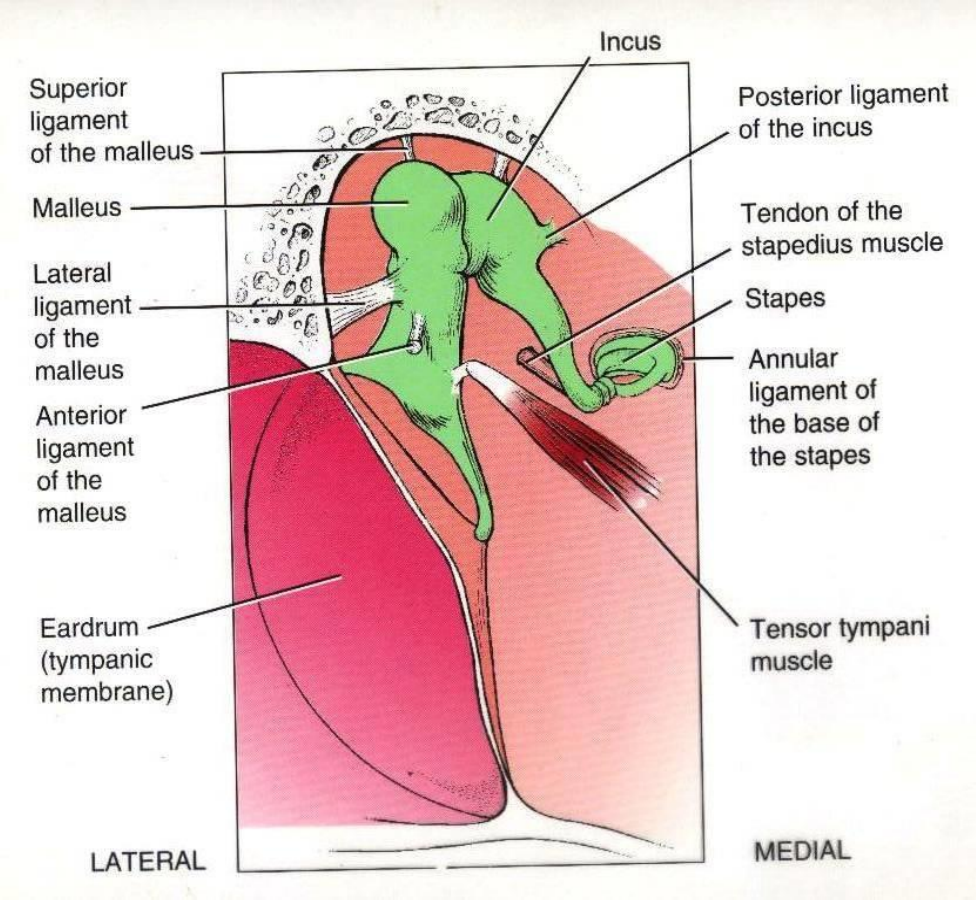
Muscles in middle ear

➢ **<u>Two muscles</u>**

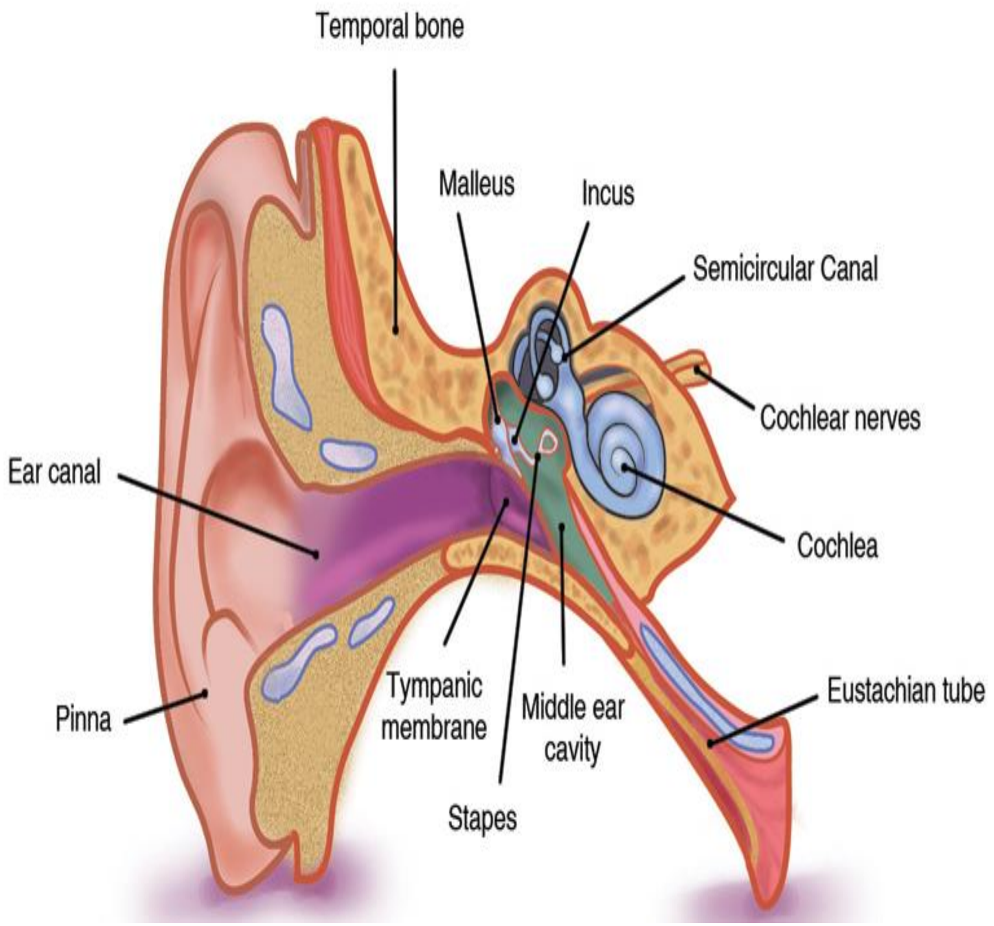
- Stapedius (supplied by facial Nerve)
- tensor tympani (supplied by Trigeminal Nerve)

➢ **<u>Ligaments of Middle Ear:</u>**

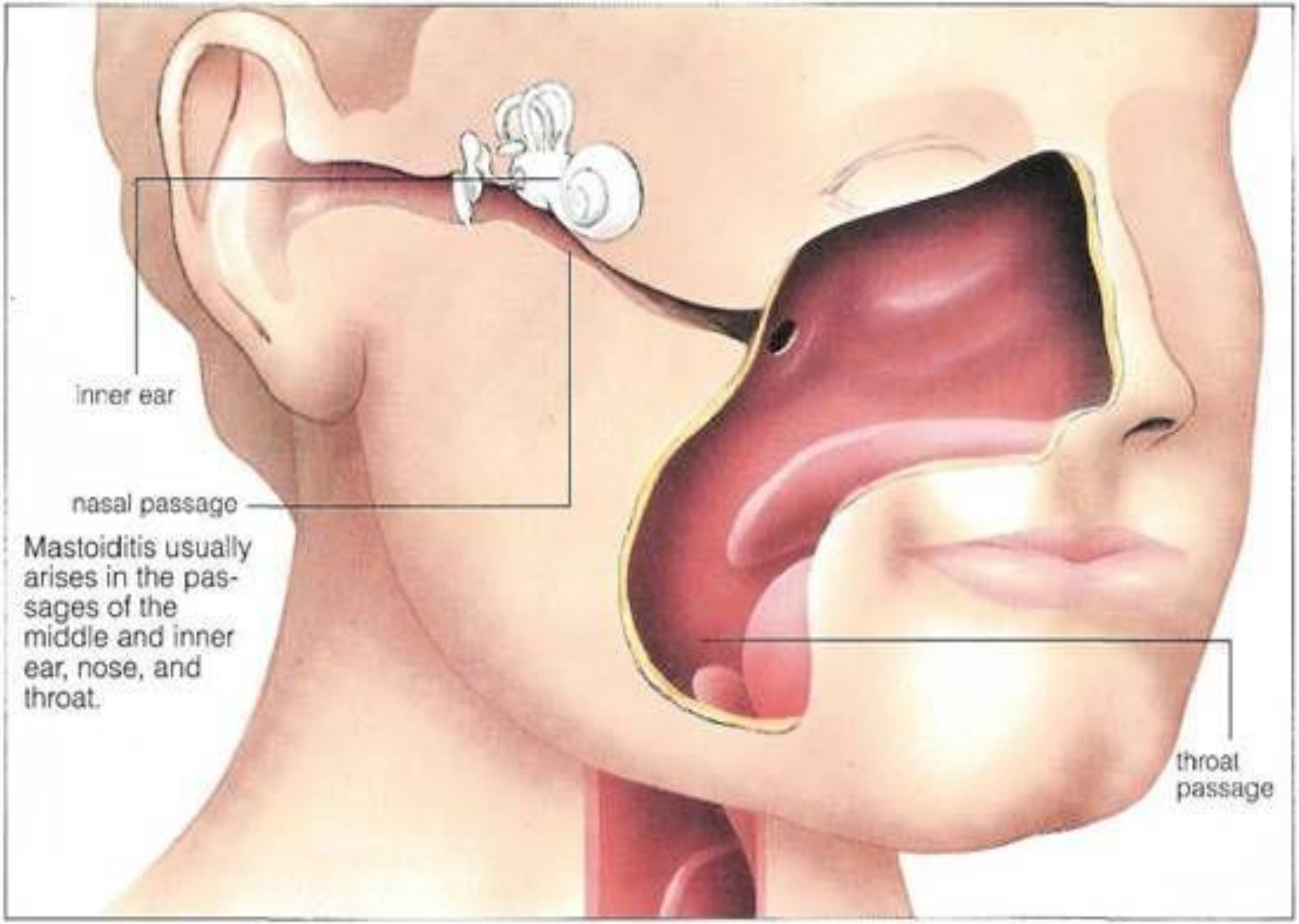
- Confine the movement of ossicles to act as a lever.
- Restrict movements to reduce the chance of damage



Eustachian Tube



- Connects middle ear to back of the throat (nasopharynx)
- Equalizes air pressure b/w middle ear and external environment
- Normally closed except during yawning, chewing or swallowing
- when fail to open –ve pressure develops so leads to stiffness in the middle ear mechanical transmission system


inner ear
nasal passage
Mastoiditis usually arises in the passages of the middle and inner ear, nose, and throat.
throat passage

Functions of middle ear

1. Transmission of sound from external ear to fluid filled inner ear thru ossicular chain.

2. Impedance matching

3. Attenuation reflex.

Impedance Matching

Match the resistance of sound wave in fluid vs. the resistance in air

What is impedance matching

- When sound is passing from one medium to the other medium of same acoustic resistance no sound energy is lost.
- But if acoustic resistance of 2 media widely differ (their impedance do not match) the amount of sound energy will be lost.

Impedance is obstacle in the pathway of sound and is expressed as amount of sound energy lost when sound passes in a medium.

The Tympanic Membrane and the Ossicular System

- Tympanic membrane functions to transmit vibrations in the air to the fluid in cochlea. So when a sound in air strikes a fluid boundary there is loss of sound energy which is equivalent to 30dB.

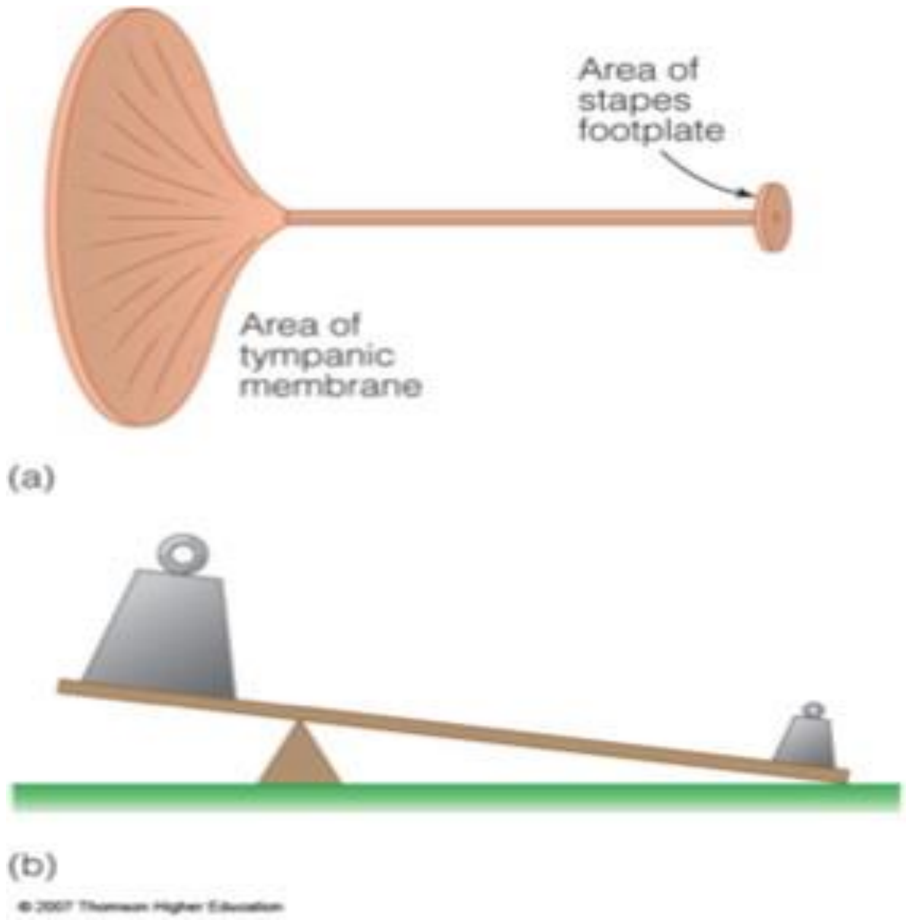
- The middle ear thus functions as impedance matching device by amplifying the sound pressure, to cause movement of fluid in Scala vestibule of cochlea

- There are 2 mechanisms by which impedance matching is obtained.

Amplification of airborne sound waves

- Amplifies the signal because the area of the tympanic membrane is 21 times larger than the oval window

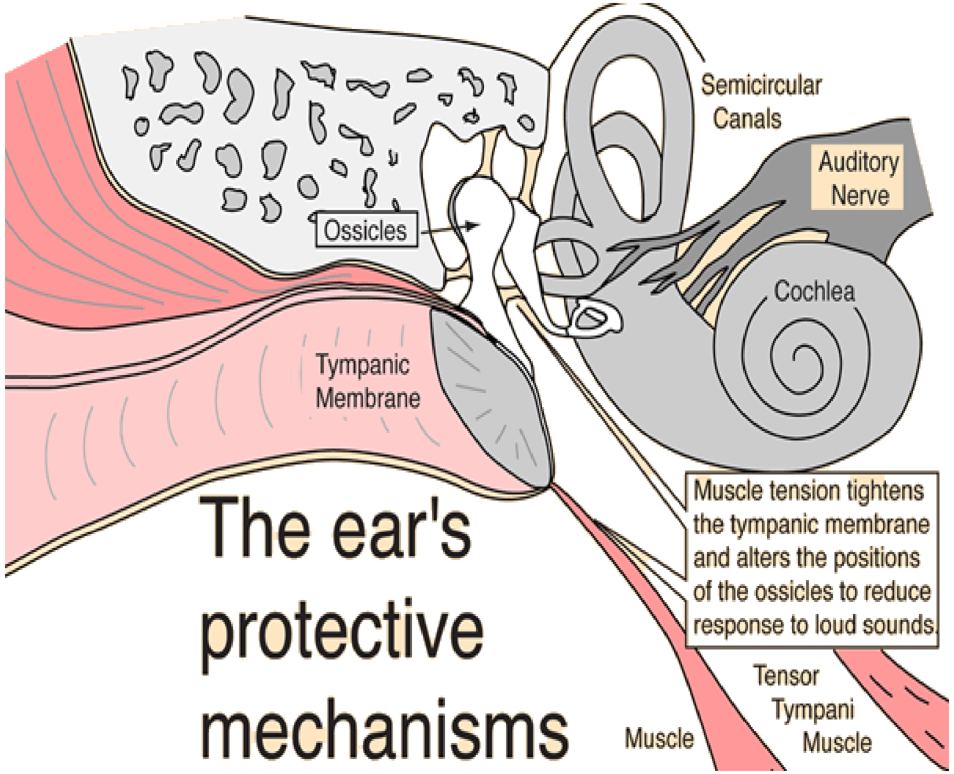
- Pressure is increased as force exerted on the tympanic membrane is conveyed by the ossicles to the oval window



Attenuation of Sound by Muscle Contraction(attenuation reflex)

- This reflex starts in response to loud sound or when the person starts to vocalize.

- A loud noise initiates reflex contraction of these muscles after (40 - 80 milliseconds) attenuates the movement of bones as ossicular system become rigid so reducing the sound transmission by 15dB



Advantages of attenuation reflex

1. It reduces the intensity of loud sound, so protects cochlea from damaging vibrations caused by excessively loud sound
2. It attenuates and mask all low frequency environment sounds & allows the person to concentrate on the sounds above 1000 cycles/sec
3. When a person is talking loudly, it decreases person’s hearing sensitivity to its own speech

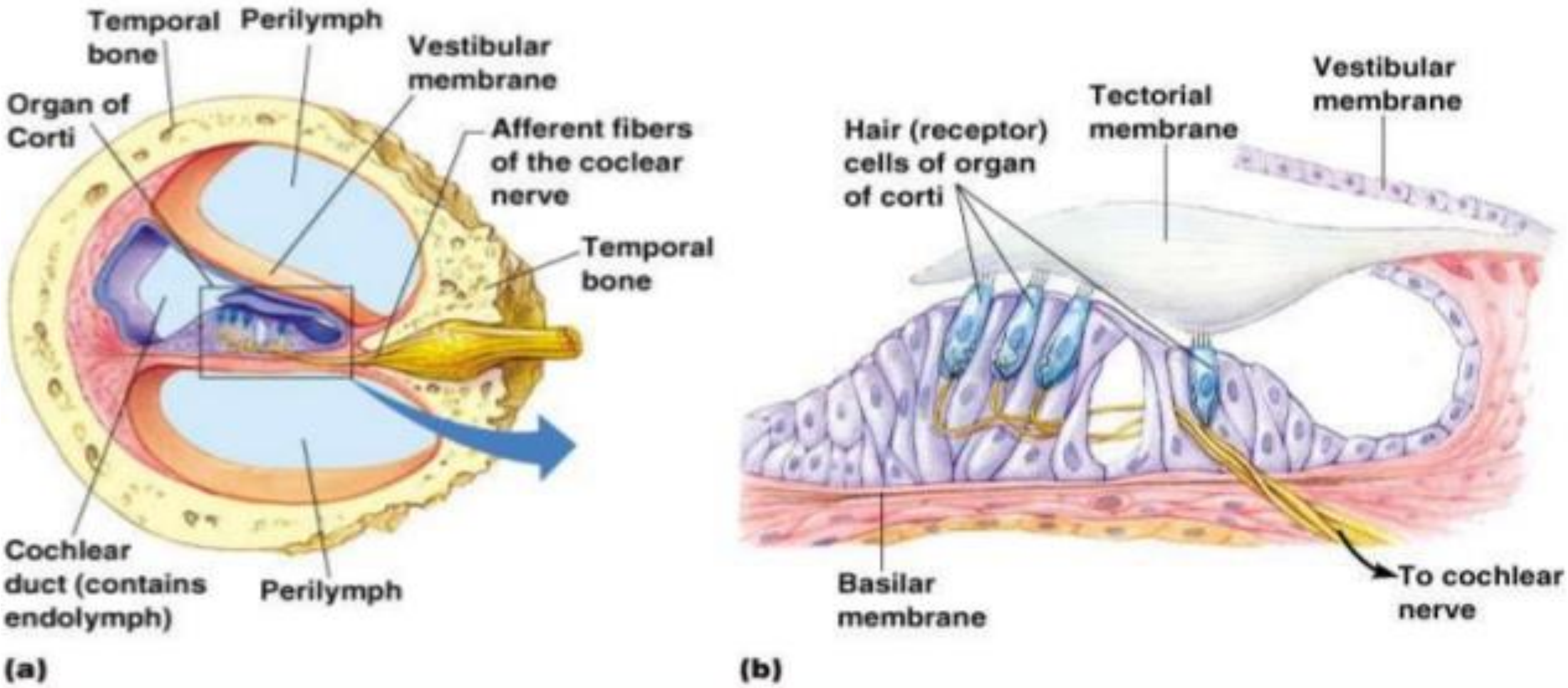
Inner ear (cochlea/ vestibule)

Cochlea:

- – spiral shaped series of 3 tubular canals or ducts.
- – contain organ of corti
- – Organ of corti contains the receptor cells and is site of auditory transduction

SENSORY CELLS FOR HEARING

- Contains Glutamate (transmitter)
- Connected with Auditory nerve fibers



ORGAN OF CORTI

- It acts as ‘sensor ` (auditory receptor organ)of cochlea.=

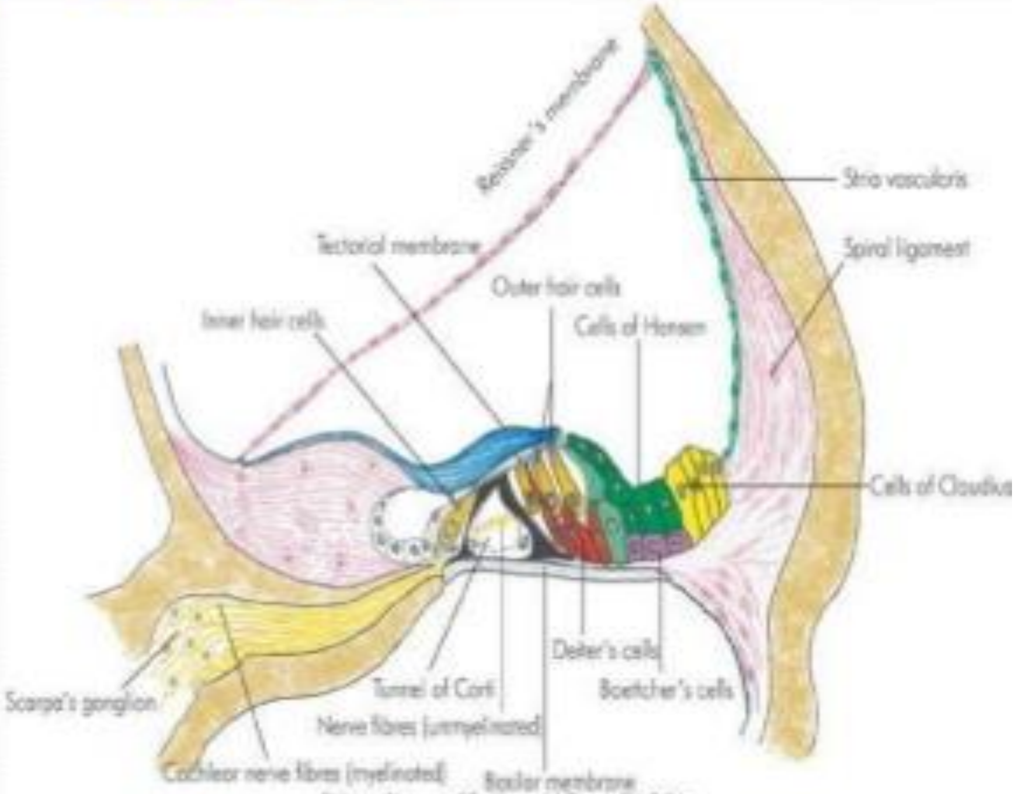
  inner hair cell

  outer hair cell

  supporting cell

  basilar membrne

  tectotial membrane

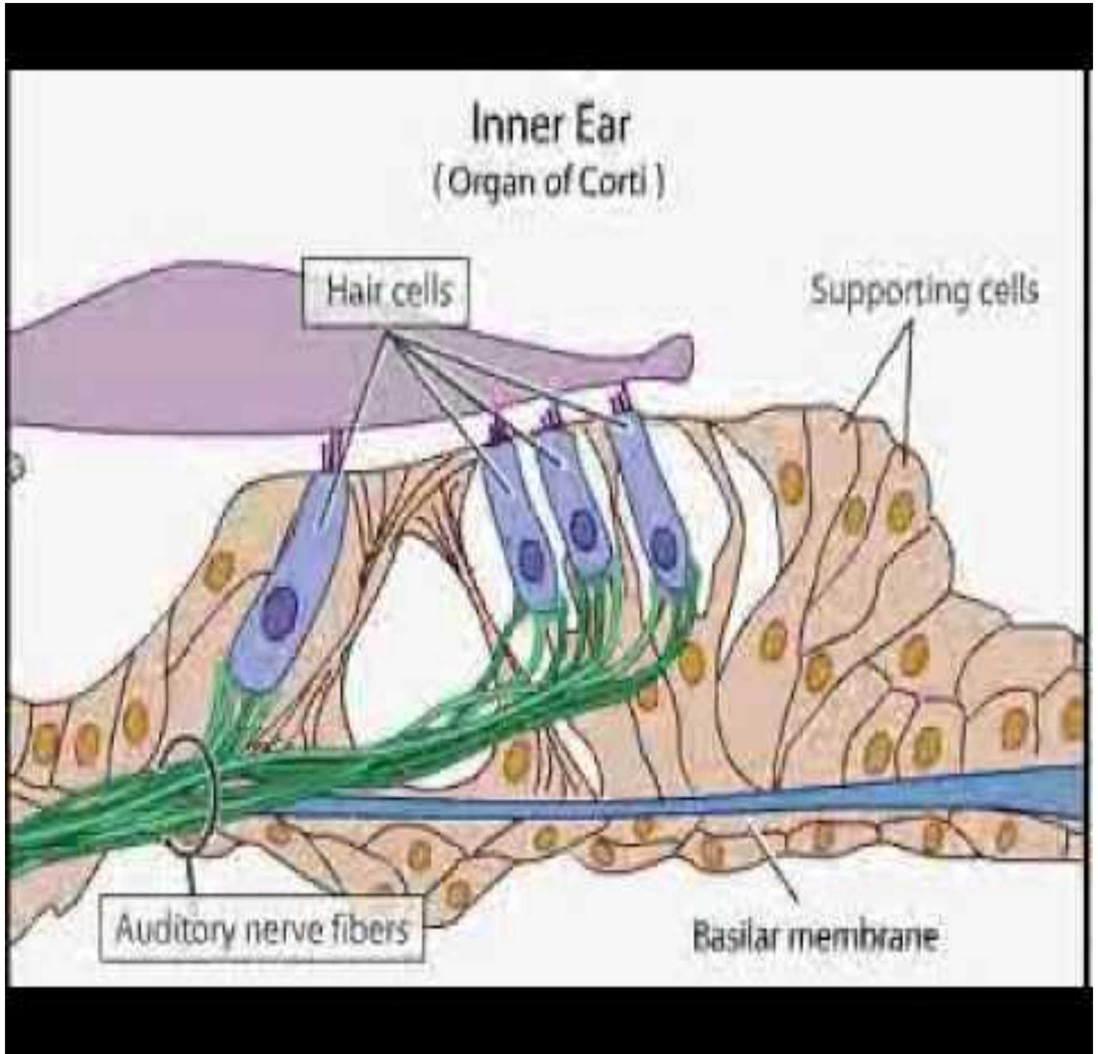


Dhingra: Diseases of Ear Nose and Throat, 5th Edition

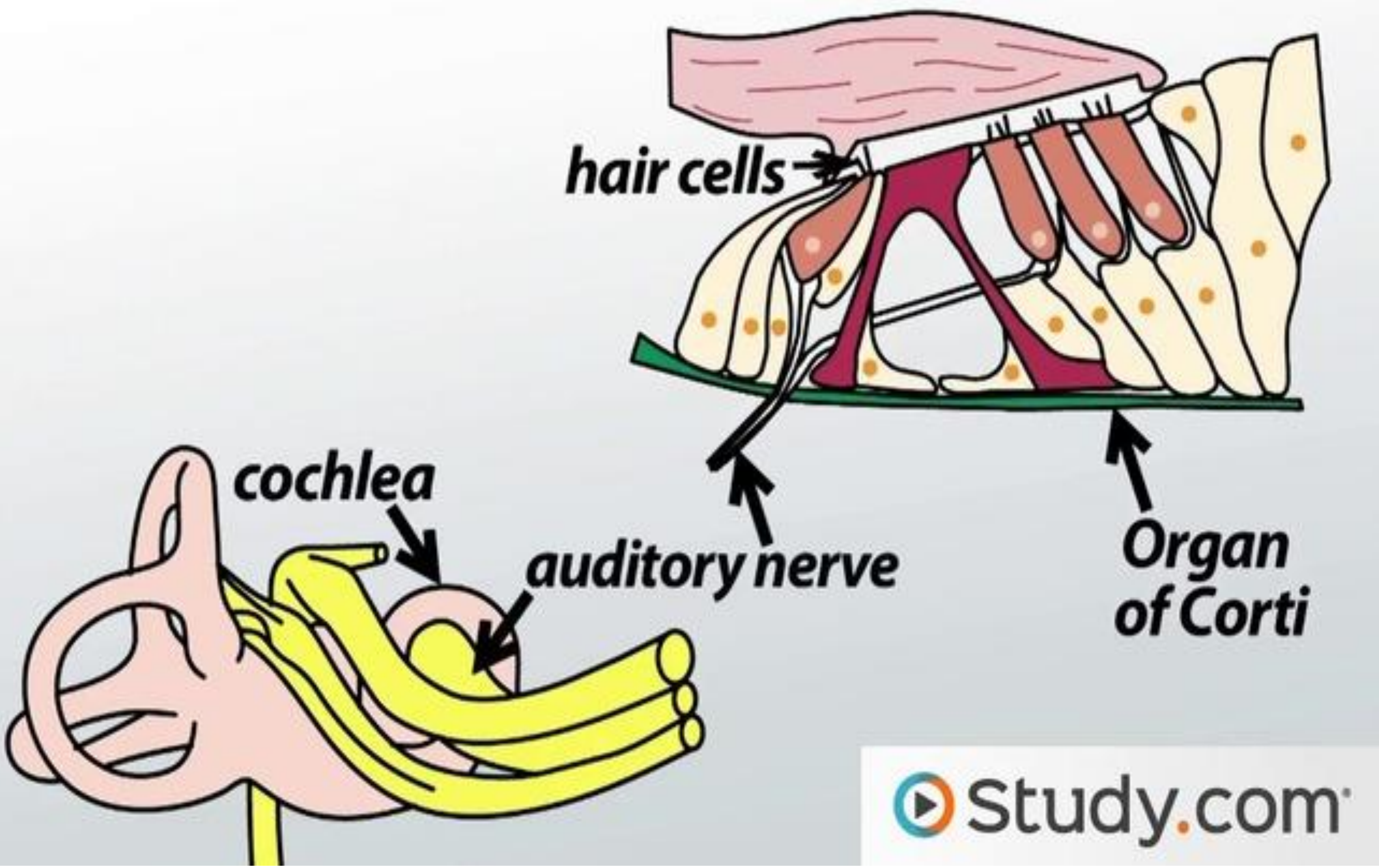
Stimulation of hair cells

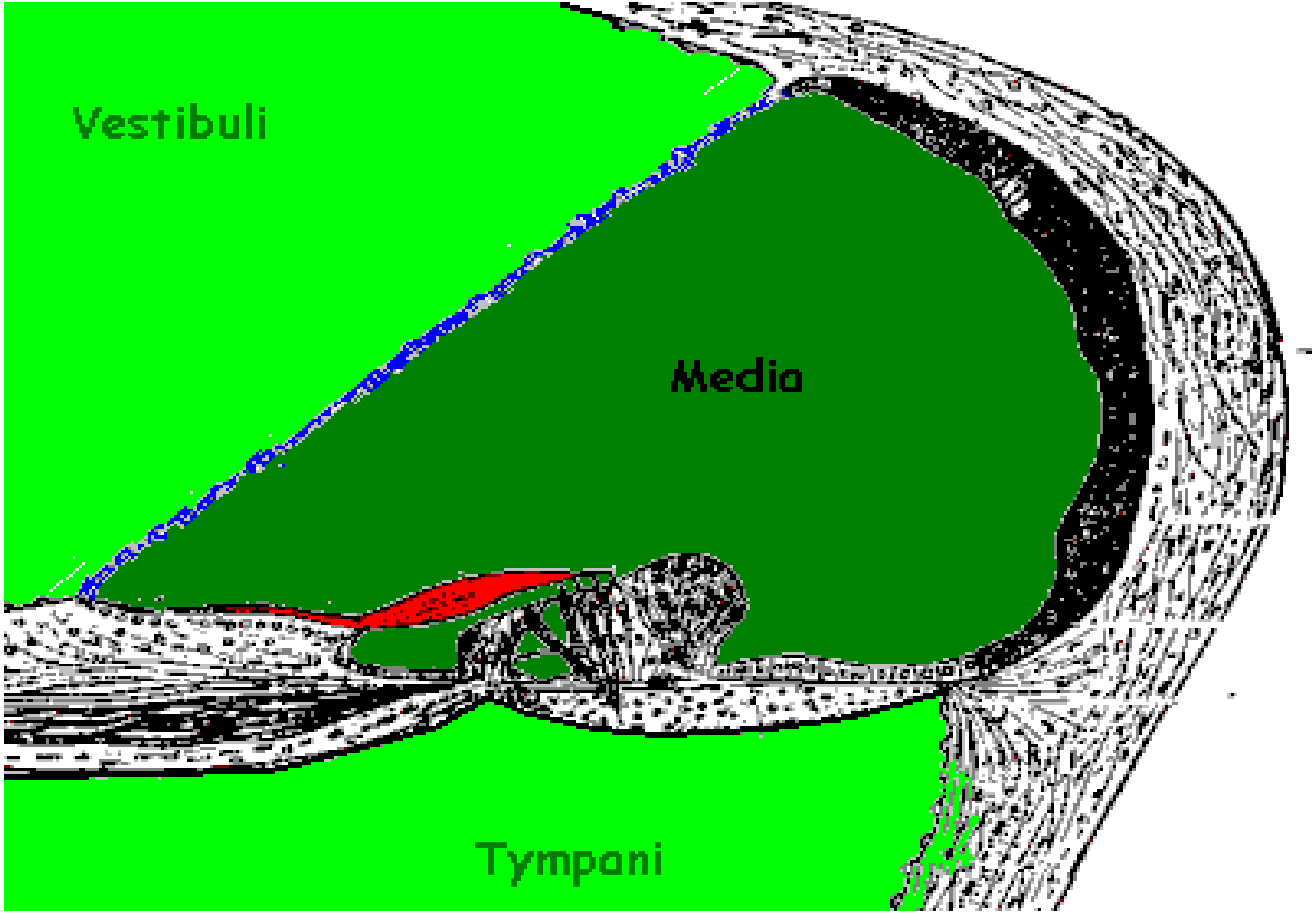
Hearing: Hair Cell Transduction

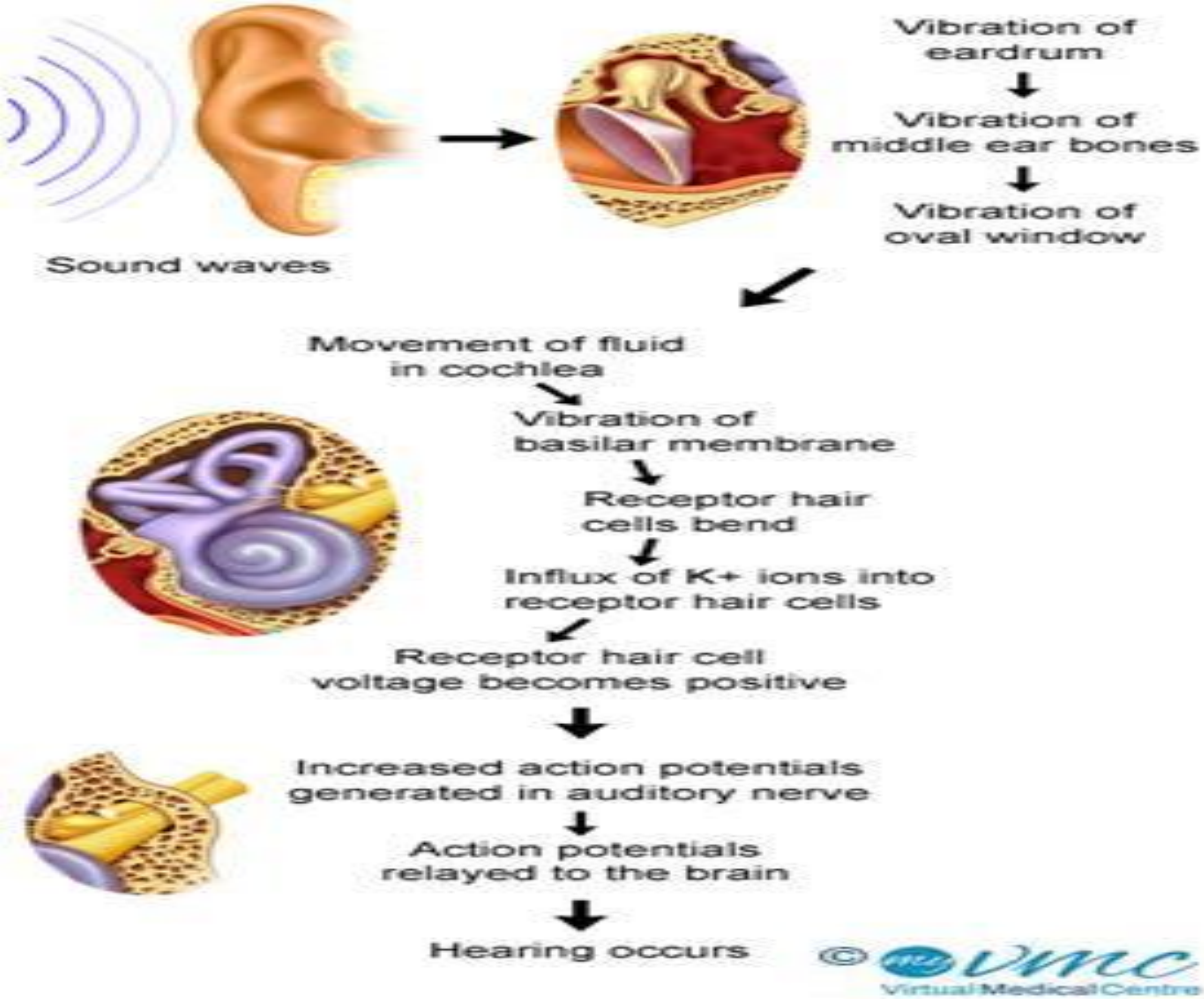
- Fluid wave moves
- Tectoral membrane
- Steriocilia move
- Ion channels open
- Depolarization
- NT release
- Sensory nerve AP



CRANIAL NERVE VIII, AUDITORY NERVE, VESTIBULOCOCHLEAR NERVE




Vestibuli
Media
Tympani


Sound waves
Vibration of eardrum
Vibration of middle ear bones
Vibration of oval window
Movement of fluid in cochlea
Vibration of basilar membrane
Receptor hair cells bend
Influx of K+ ions into receptor hair cells
Receptor hair cell voltage becomes positive
Increased action potentials generated in auditory nerve
Action potentials relayed to the brain
Hearing occurs
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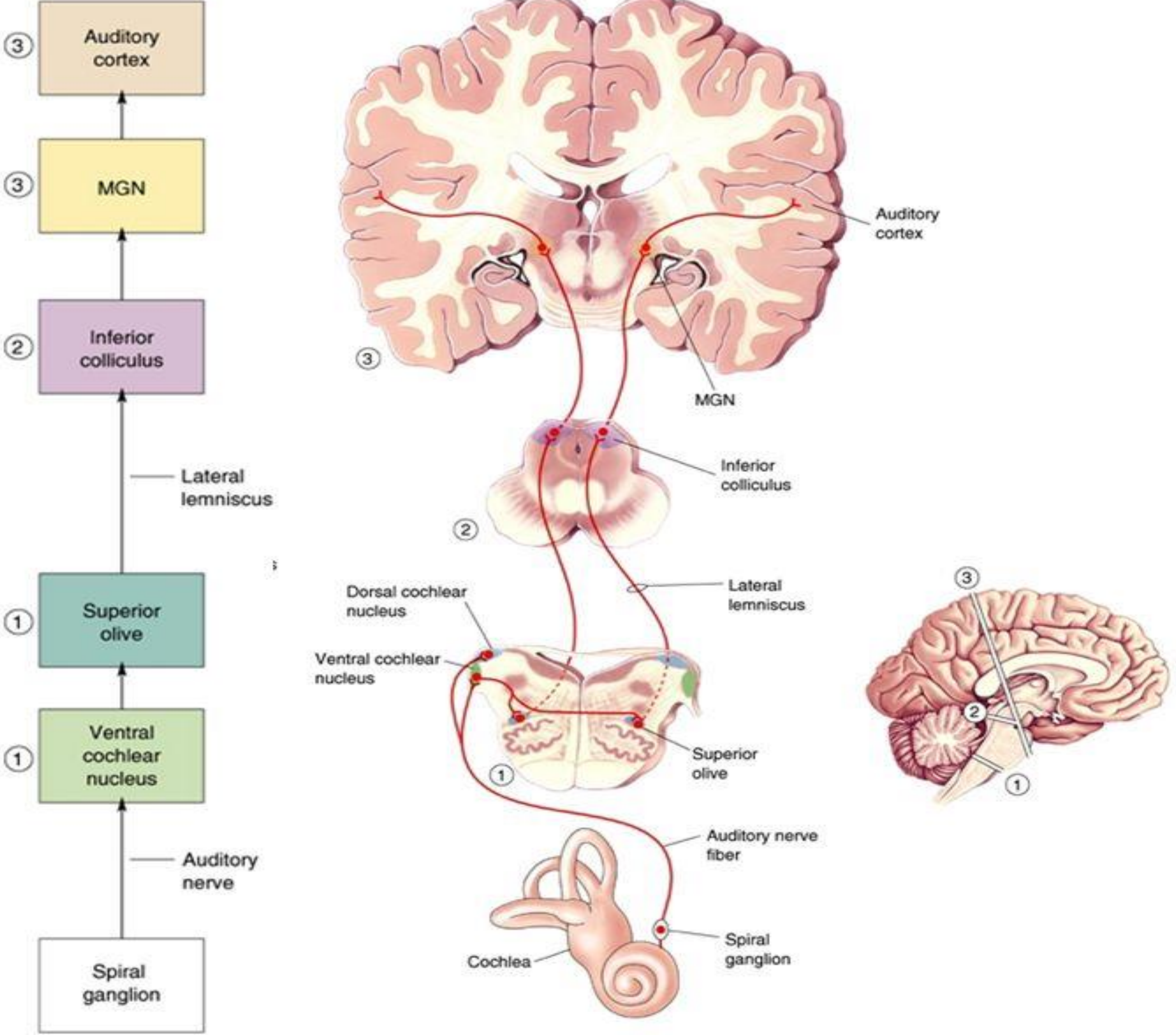
Physiology of Hearing:

- 1) Sound waves strike **tympanic membrane** causing it to vibrate. Membrane moves in with each compression & out with each decompression.

- 2) Movement of membrane is transferred to the **ossicles** in order; **malleus, incus, & stapes**. Vibrations are conducted across tympanic cavity and amplified about 22 times.
- 3) The stapes moves in and out at the **oval window**. Since liquids don't compress, the pressure of the **perilymph** in the **scala vestibuli** (vestibular duct) goes up and down. The **secondary tympanic membrane** over the **round window** bulges out with each increase in pressure.
- 4) Each increase in pressure in the scala vestibuli pushes the **vestibular membrane** down, which increases the pressure of the **endolymph** in the **cochlear duct**, which pushes the **basilar membrane** down.
- 5) Vibration of the basilar membrane causes vibration of hair cells against the **tectorial membrane**. This causes movement of the **stereocilia** on the hair cells.
- 6) Information about the region of the cochlea (pitch) & intensity of the stimulus are relayed to CNS over cochlear branch of vestibulocochlear cranial nerve.

Auditory Pathway



The Ascending Auditory Pathway

Primary Auditory Cortex

Medial Geniculate Body

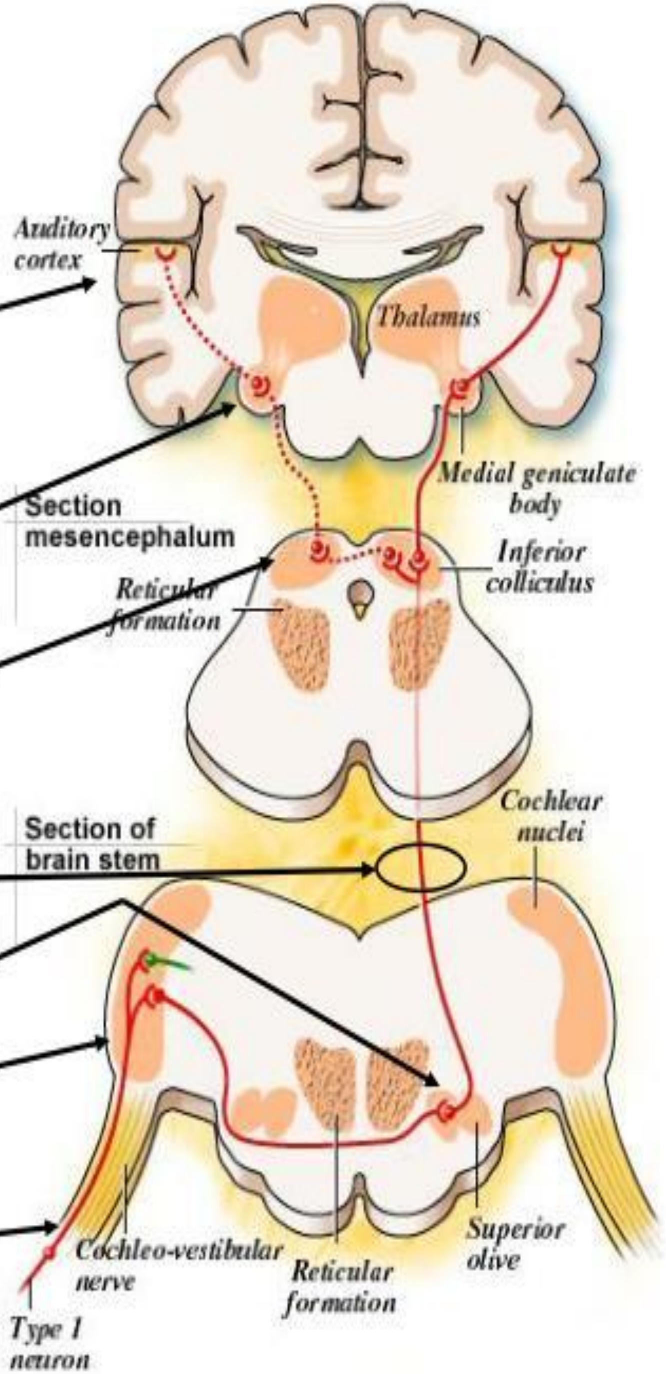
Inferior Colliculus

Lateral Lemniscus

Superior Olivary Complex

Cochlear Nucleus

VIIIth Nerve



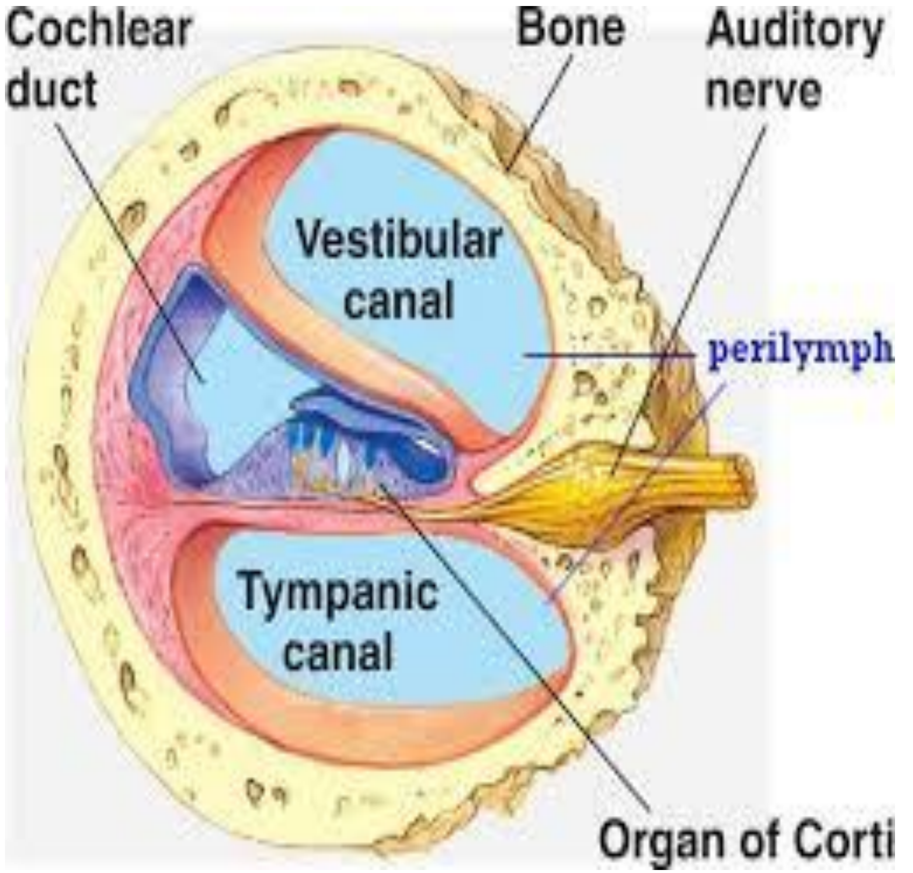
Determination of loudness

- With louder sound, amplitude of vibration of the basilar membrane and hair cells increase
- Stimulation of more and more hair cells
- Stimulation of outer hair cells

- Impedence matching is the function of inner ear
- Attanuation reflex is a destructive reflex
- Tympanic membrane has a diameter less than oval window
- Speed of sound waves is faster is fluid medium
- Organ of corti present on reissiener membrane
- First order neurons present in inferior coliculus in auditory pathway
- Rhythm/music interpretation is associated with primary auditory cortex
- Lesion in wernikies area causes hearing loss

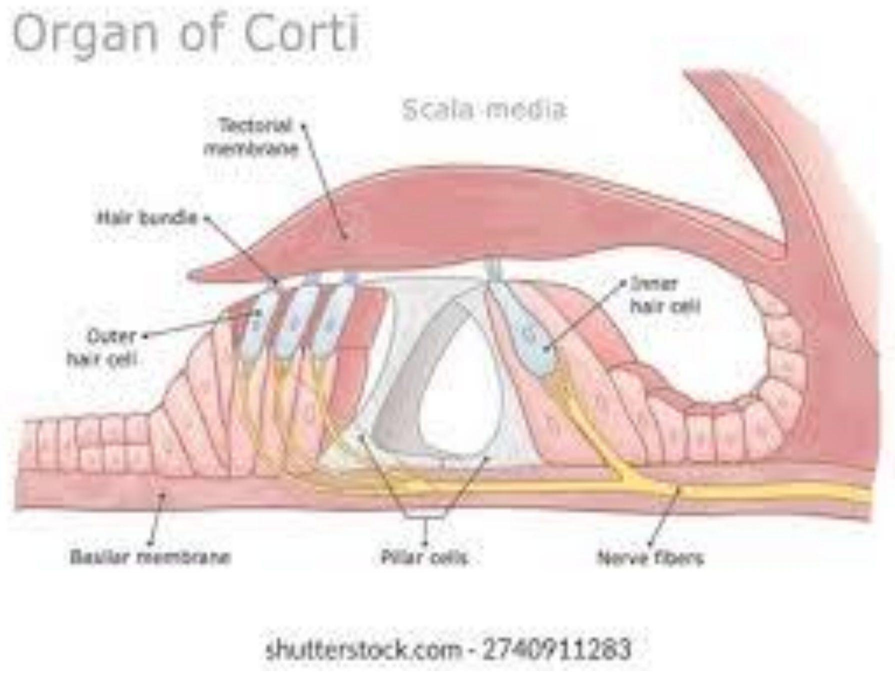
The cochlea contains the organ of Corti, the sense organ for hearing

- The pea-sized, snail-shaped cochlea, the “hearing” portion of the inner ear, is a coiled tubular system lying deep within the temporal bone) (cochlea means “snail”).
- It is easier to understand the functional components of the cochlea by “uncoiling” it
- The cochlea is divided throughout most of its length into three fluid-filled longitudinal compartments. A blind-ended cochlear duct, which is also known as the scala media, constitutes the middle compartment.
- The upper compartment, the scala vestibule, the scala tympani, the lower compartment.
- The fluid within the scala vestibuli and scala tympani is called perilymph.
- The cochlear duct contains a slightly different fluid, the endolymph
- The region beyond the tip of the cochlear duct where the fluid in the upper and lower compartments is continuous is called the helicotrema.
- The scala vestibuli is sealed from the middle ear cavity by the oval window, to which the stapes is attached.
- Another small membrane-covered opening, the round window, seals the scala tympani from the middle ear.
- The thin vestibular membrane forms the ceiling of the cochlear duct and separates it from the scala vestibuli.
- The basilar membrane forms the floor of the cochlear duct, separating it from the scala tympani.
- The basilar membrane is especially important because it bears the organ of Corti, the sense organ for hearing.



Hair cells in the organ of Corti transduce fluid movements into neural signals.

- The organ of Corti, which rests on top of the basilar membrane throughout its full length, contains auditory hair cells that are the receptors for sound.
- The 15,000 hair cells within each cochlea are arranged in four parallel rows along the length of the basilar membrane:
- one row of inner hair cells and three rows of outer hair cells.
- Protruding from the surface of each hair cell are about 100 hairs known as stereocilia, which are actin-stiffened microvilli
- Hair cells are mechanoreceptors; they generate neural signals when their surface hairs are mechanically deformed by fluid movements in the inner ear.
- These stereocilia contact the tectorial membrane, an awning like projection overhanging the organ of Corti throughout its length



- The piston like action of the stapes against the oval window sets up pressure waves in the upper compartment.
- pressure is dissipated in two ways as the stapes causes the oval window to bulge inward:

(1) displacement of the round window

- In the first of these pathways, the pressure wave pushes the perilymph forward in the upper compartment, around the helicotrema, and into the lower compartment, where it causes the round window to bulge outward into the middle ear cavity to compensate for the pressure increase.
- As the stapes rocks backward and pulls the oval window outward toward the middle ear, the perilymph shifts in the opposite direction, displacing the round window inward.
- This pathway does not result in sound reception; it just dissipates pressure.

(2) deflection of the basilar membrane

- Pressure waves of frequencies associated with sound reception take a “shortcut”.
- Pressure waves in the upper compartment are transferred through the thin vestibular membrane, into the cochlear duct and then through the basilar membrane into the lower compartment.
- Transmission of pressure waves through the basilar membrane causes this membrane to move up and down, or vibrate, in synchrony with the pressure wave.
- Because the organ of Corti rides on the basilar membrane, the hair cells also move up and down.

Role of The Inner Hair Cells

- The inner and outer hair cells differ in function. The inner hair cells are the ones that “hear”:
- They transform the mechanical forces of sound (cochlear fluid vibration) into the electrical impulses
- Because the stereocilia of these receptor cells contact the stiff, stationary tectorial membrane, they are bent back and forth when the oscillating basilar membrane shifts their position in relationship to the tectorial membrane
- This back-and-forth mechanical deformation of the hairs alternately opens and closes mechanically gated cation channels in the hair cell, resulting in alternating depolarizing and hyperpolarizing potential changes—the receptor potential— at the same frequency as the original sound stimulus.
- The stereocilia of each hair cell are organized into rows of graded heights ranging from short to tall in a precise pattern resembling organ pipes.
- Tip links, which are CAMs (cell adhesion molecules, link the tips of stereocilia in adjacent rows.
- When the basilar membrane moves upward, the bundle of stereocilia bends toward its tallest membrane, stretching the tip links.
- Stretched tip links open the cation channels to which they are attached . The resultant ion movement is unusual because of the unique composition of the endolymph that bathes the stereocilia.
- In sharp contrast to ECF elsewhere, endolymph has a higher concentration of K1 than found inside the hair cell.
- When more cation channels are pulled open, more K1 enters the hair cell. This K1 depolarizes the hair cell . When the basilar membrane moves in the opposite direction, the hair bundle bends away from the tallest stereocilium, relief the tip links and closing all the channels. As a result, K1 entry ceases, hyperpolarizing the hair cell
- Depolarization of these hair cells opens more voltage-gated Ca21 channels. increases their rate of neurotransmitter secretion, which steps up the rate of firing in the afferent fibers with which the inner hair cells synapse.

Role of The Outer Hair Cells

- the outer hair cells do not signal the brain about incoming sounds.
- Instead, the outer hair cells actively and rapidly change length in response to changes in membrane potential, a behavior known as electromotility.
- the outer hair cells in some way control the sensitivity of the inner hair cells at different sound pitches, a phenomenon called "tuning" of the receptor system.
- In support of this concept, a large number of retrograde nerve fibers pass from the brain stem to the vicinity of the outer hair cells.
- Stimulating these nerve fibers can actually cause shortening of the outer hair cells and possibly also change their degree of stiffness.
- These effects suggest a retrograde nervous mechanism for control of the ear's sensitivity to different sound pitches, activated through the outer hair cells

Pitch discrimination depends on the region of the basilar membrane that vibrates.

- Pitch discrimination (that is, the ability to distinguish among various frequencies of incoming sound waves) depends on the shape and properties of the basilar membrane, which is narrow and stiff at its oval window end and wide and flexible at its helicotrema end.
- Different regions of the basilar membrane naturally vibrate maximally at different locations.
- The narrow end nearest the oval window vibrates best with high-frequency pitches, whereas the wide end nearest the helicotrema vibrates maximally with low-frequency tones.
- The pitches in between are sorted out precisely along the length of the membrane from higher to lower frequency.
- Overtones of varying frequencies cause many points along the basilar membrane to vibrate simultaneously but less intensely than the fundamental tone, enabling the CNS to distinguish the timbre of the sound (timbre discrimination).

Loudness discrimination depends on the amplitude of vibration.

- Intensity (loudness) discrimination depends on the amplitude of vibration.
- As sound waves originating from louder sound sources strike the eardrum, they cause it to vibrate more vigorously (that is, bulge in and out to a greater extent) but at the same frequency as a softer sound of the same pitch.
- The greater tympanic membrane deflection translates into greater basilar membrane movement in the region of peak responsiveness, causing greater bending of the hairs in this region.
- The CNS interprets this greater hair bending as a louder sound.
- Thus, pitch discrimination depends on “where” the basilar membrane maximally vibrates and loudness discrimination depends on “how much” this place vibrates.

Thank you